



In vitro screening for aryl hydrocarbon receptor agonistic activity in 200 pesticides using a highly sensitive reporter cell line, DR-EcoScreen cells, and *in vivo* mouse liver cytochrome P450-1A induction by propanil, diuron and linuron

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ABSTRACT

The aryl hydrocarbon receptor (AhR) is a ligand-dependent transcription factor that regulates genes involved in xenobiotic metabolism, cellular proliferation and differentiation. In this study, we have developed a highly sensitive AhR-mediated reporter cell line, DR-EcoScreen cells, which are mouse hepatoma Hepa1c1c7 cells stably transfected with a reporter plasmid containing seven copies of dioxin-responsive element. Using these DR-EcoScreen cells, we performed the reporter gene assay and characterized the AhR agonistic activities of 200 pesticides (29 organochlorines, 11 diphenyl ethers, 56 organophosphorus pesticides, 12 pyrethroids, 22 carbamates, 12 acid amides, 7 triazines, 6 ureas, and 45 others). Eleven of the 200 pesticides (acifluorfen-methyl, bifenoxy, chlorpyrifos, isoxathion, quinalphos, chlorpropham, diethofencarb, propanil, diuron, linuron, and prochloraz) showed AhR-mediated transcriptional activity. In particular, three herbicides (propanil, diuron, and linuron) have a common chemical structure and showed more potent agonistic activity than other pesticides. To investigate the *in vivo* effects, we examined the gene expression of AhR-inducible cytochrome P450 1As (CYP1As) in the liver of female C57BL/6 mice intraperitoneally injected with these three herbicides (≤ 300 mg kg⁻¹) by quantitative RT-PCR, resulting in induction of significant high levels of CYP1A1 and CYP1A2 mRNAs. This indicates that propanil, diuron and linuron possess AhR-mediated transactivation effect *in vivo* as well as *in vitro*. Through the present study, we demonstrated that DR-EcoScreen cells are useful for sensitive, rapid and simple identification of AhR agonists among a large number of environmental chemicals.

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1. Introduction

The aryl hydrocarbon receptor (AhR) is a member of the superfamily of basic helix-loop-helix (bHLH)/Per-Arnt-Sim (PAS) transcription factors, which activates gene expression in a ligand-dependent manner (Hankinson, 1995; Swanson and Bradfield, 1993). The AhR was initially identified as an intracellular receptor for 2,3,7,8-tetrachlorodibenzo-*p*-dioxin (TCDD) and halogenated aromatic pollutants that are structurally related to TCDD (Poland et al., 1976; Poland and Knutson, 1982). Unbound AhRs exist in the cytosolic compartment as a multiprotein complex, and can bind to both exogenous and endogenous ligands. The AhR bound to ligand (ligand–AhR complex) translocates from the cytoplasm to the nucleus, where it further binds to its dimerization partner, AhR nuclear translocator (ARNT), which is another bHLH/PAS tran-

scription factor. The activated AhR/ARNT heterodimer complex binds to its cognate DNA sequences, termed dioxin-responsive elements (DREs) (5′-TNGCGTG-3′), to activate expression of AhR target genes, among which the cytochrome P4501A1 (CYP1A1) gene is best characterized (Hankinson, 1995; Reyes et al., 1992). Many toxicological studies have reported that the interaction of AhR ligands with this receptor is involved in a wide variety of species- and tissue-specific, toxic and biological effects, such as birth defects, immunotoxicity, neurotoxicity, lethality, tumor promotion, and enzyme induction (Poland and Knutson, 1982; Safe, 1986; Fernandez-Salguero et al., 1996; Mimura et al., 1997).

The ligands for AhR have been categorized into two major groups (Denison and Nagy, 2003). One is a classical group having high affinity for AhR. Some members of this group are recognized as environmental contaminants such as halogenated aromatic hydrocarbons (HAHs) including TCDD, polycyclic aromatic hydrocarbons (PAHs) including 3-methylcholanthrene (3-MC), and related compounds (Poland and Knutson, 1982; Safe, 1990; Kafafi

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et al., 1993). The other group is a nonclassical group, which consists of structurally diverse chemicals that are relatively weak inducers of *CYP1A1* and/or have quite lower affinity for AhR than TCDD. These AhR ligands are considered to probably have a broad spectrum of their chemical structures, and widely exist in our environment.

The ubiquitous nature of pesticide usage with minimal precautions has resulted in contamination of food, the workplace, and natural environment with them. Some pesticides have been recently reported to possess weak AhR agonistic activity. Long et al. (2003) reported that eight of 23 pesticides currently used in Denmark revealed AhR-mediated activity in human and rat hepatoma cell lines. In addition, a carbamate-type insecticide, carbaryl (Ledirac et al., 1997; Denison et al., 1998), and a urea-type herbicide, diuron (Noguerol et al., 2006; Zhao et al., 2006), were found to act as an AhR agonists in AhR-mediated reporter gene assays using mammalian hepatoma cells or recombinant yeast cells. Although these pesticides are relatively weak inducers or AhR ligands when compared with TCDD, these data are of particular importance due to the fact that pesticides are intentionally released in a large quantity into the environment. Moreover, ability of other numerous pesticides to activate AhR is not fully elucidated. Therefore, it is important issue to find whether a large number of pesticides act as AhR activators.

We have previously investigated 200 pesticides including related isomers and metabolites regarding the estrogenicity via human estrogen receptor (hER) subtypes, hER α and hER β , and the androgenicity via a human androgen receptor (hAR), by *in vitro* reporter gene assays to show that a variety of pesticides act as ER α / β agonists and/or AR antagonists (Kojima et al., 2004). In addition, our recent paper provided evidence that three of these 200 pesticides possess *in vitro* peroxisome proliferator-activated receptor (PPAR) α agonistic activity (Takeuchi et al., 2006). Therefore, some of these 200 pesticides may possibly interact with other receptor families such as AhR in addition to ER α / β , AR and PPAR α .

In the present study, we have developed a new, sensitive and rapid reporter gene assay using a genetically engineered stable cell line, designated as DR-EcoScreen to detect AhR agonistic activity of a large number of environmental chemicals. Then, we used this DR-EcoScreen reporter gene assay to determine whether 200 pesticides have AhR agonistic activities. Furthermore, we examined *in vivo* effects of some of these pesticides based on expression induction of *CYP1A1* and *CYP1A2* genes via AhR (Dey et al., 1999; Negishi and Nebert, 1979). Thus, we here provide the evidence that eleven pesticides, among the 200 pesticides tested, have *in vitro* AhR agonistic activity, and particularly, three herbicides (propanil, diuron and linuron) are also *in vivo* inducers of *CYP1A* genes in the mouse liver.

2. Materials and methods

2.1. Chemicals

The 200 pesticides tested are listed in Table 1. These pesticides had a purity of 95–100% and were purchased from Wako Pure Chemical Industries Ltd. (Wako; Osaka, Japan), Sigma–Aldrich (St. Louis, MO, USA), Dr. Ehrenstorfer GmbH (Augsburg, Germany), AccuStandard Inc. (New Haven, CT, USA), and Hayashi Pure Chemical Industries Ltd. (Osaka, Japan). TCDD, 3-MC, benzo[α]pyrene, β -naphthoflavone, and dimethylsulfoxide (DMSO) were purchased from Wako. 3,3',4,4',5-Pentachlorobiphenyl (PCB # 126) was purchased from AccuStandard Inc. Chloropyrifos-oxon and isoxathion-oxon, metabolites of chloropyrifos and isoxathion respectively, were purchased from Wako.

Table 1

The 200 pesticides tested in the reporter gene assays for AhR

Compound
I Organochlorines (<i>n</i> = 29)
Aldrin
α -BHC
β -BHC
γ -BHC
δ -BHC
Captan
<i>cis</i> -Chlordane
<i>trans</i> -Chlordane
Chlorobenzilate
Chloropropylate
Chlorothalonil
<i>o,p'</i> -DDT
<i>p,p'</i> -DDT
<i>p,p'</i> -DDE
<i>p,p'</i> -DDD
Dichlobenil
Dicofol
Dieldrin
α -Endosulfan
β -Endosulfan
Endosulfan sulfate
Endrin
Folpet
Fthalide
Heptachlor
Heptachlor epoxide
Methoxychlor
Pentachlorophenol
Quintozone
II Diphenyl ethers (<i>n</i> = 11)
Acifluorfen
Acifluorfen-methyl
Bifenox
Chlormethoxyfen
Chlornitrofen
CNP-amino
Chloroxurone
Diclofop-methyl
Fluazifop-butyl
Nitrofen
Oxyfluorfen
III Organophosphorus (<i>n</i> = 56)
Acephate
Anilofos
Bromophos-ethyl
Bromophos-methyl
Butamifos
Chlorpyrifos
Chlorpyrifos-methyl
Cyanofenphos
Cyanophos
Diazinon
Dichlofenthion
Dichlorvos
Dimethoate
Dioxabenzofos
Disulfoton
EPN ^a
Edifenphos
Ethion
Ethoprophos
Fenamiphos
Fenchlorphos
Fenitrothion
Fenitrothion oxon
Fensulfothion
Fenthion
Glyphosate
Iprobenfos
Isofenphos
Isoxathion
Leptophos
Malathion
Mecarbam

(continued on next page)

Table 1 (continued)

Compound
Methamidophos
Methidathion
Methyl-parathion
Monocrotophos
Parathion
Phenthoate
Phorate
Phosalone
Phosmet
Piperophos
Pirimiphos-methyl
Profenofos
Propaphos
Prothiofos
Prothiofos oxon
Pyridaphenthion
Quinalphos
Terbufos
Tetrachlorvinphos
Thiometon
Tolclofos-methyl
Tolclofos-methyl oxon
Trichlorfon
Vamidothion
IV Pyrethroids (<i>n</i> = 12)
Cyfluthrin
Cyhalothrin
Cypermethrin
Deltamethrin
Etofenprox
Fenvalerate
Flucythrinate
Fluvalinate
Permethrin
Pyrethrin
Tefluthrin
Tralomethrin
V Carbamates (<i>n</i> = 22)
Bendiocarb
Benomyl
Carbaryl
Carbendazim
Carbofuran
Chlorpropham
Diethofencarb
Dimepiperate
Esprocarb
Ethiofencarb
Fenobucarb
Isoprocarb
Methiocarb
Methomyl
Molinate
Oxamyl
Phenmedipham
Pirimicarb
Pyributicarb
Thiobencarb
Thiobencarb sulfon
Thiram
VI Acid amides (<i>n</i> = 11)
Alachlor
Asulam
Cafenstrole
Flutolanil
Mefenacet
Mepronil
Metalaxyl
Metolachlor
Pretilachlor
Propyzamide
Thenylchlor
VII Triazines (<i>n</i> = 7)
Anilazine

Table 1 (continued)

Compound
Atrazine
Metribuzin
Prometon
Prometryn
Simazine
Simetryn
VIII Ureas (<i>n</i> = 8)
Bensulfuron-methyl
Daimuron
Diiflubenuron
Diuron
Linuron
Pencycuron
Prochloraz
Propanil
IX Others (<i>n</i> = 44)
Amitraz
Benfuresate
Bentazone
Benzoximate
Biphenyl
Bitertanol
Bromopropylate
Chinomethionat
Chloridazon
Dazomet
Diquat
Ethoxyquin
Fenarimol
Ferimzone
Fluazinam
Imazalil
Imidacloprid
Iminoctadine
Indanofan
Ioxynil
Iprodione
Isoprothiolane
Lenacil
MCPA ^b
2,4-D ^c
Paraquat
Pendimethalin
2-Phenylphenol
Probenazole
Procymidone
Propiconazole
Pyrazolynate
Pyrazoxyfen
Pyroquilon
Sethoxydim
Thiabendazole
Thiocyclam
Thiophanate-methyl
Triadimefon
Tricyclazole
Triflumizole
Trifluralin
Triforine
Vinclozolin

^a O-ethyl O-p-nitrophenyl phenylphosphonothioate.^b 4-Chloro-o-toloxycetic acid.^c 2,4-Dichlorophenoxyacetic acid.

2.2. Cell line and cell culture materials

Mouse hepatoma Hepa1c1c7 cells were obtained from the American Type Culture Collection. Fetal bovine serum (FBS) was obtained from JRH Bioscience, Inc. (Lenexa, KS, USA). Charcoal-dextran treated FBS (CD-FBS) was obtained from Hyclone (Logan, UT, USA). Alpha-modified Eagle's minimum essential medium α -MEM) was obtained from MP Biomedicals (Germany). Hygromycin was obtained from Invitrogen (San Diego, CA, USA). Glutamine

solution and penicillin–streptomycin (antibiotics) solution were obtained from Dainippon Pharmaceutical Co. Ltd. (Osaka, Japan). 0.25% Trypsin/0.02% ethylenediamine tetra-acetic acid (EDTA) disodium salt solution was obtained from Life Technologies (Paisley, UK). The luciferase substrate, Steady-Glo[™] reagent and luciferin, were purchased from Promega (Madison, WI, USA). FuGENE[®]6 Transfection Reagent was purchased from Roche Diagnostics Corp. (Indianapolis, IN, USA).

2.3. Construction of reporter plasmid, pIND-GCDR7

We cloned and amplified the regulatory element containing DREs of the *CYP1A1* gene from the genome of C57BL/6 strain mouse by PCR (Pyrobest TAKARA, Japan) using three following primers. Small letters represent flanking site sequences of the primers, and italics represent sequences of restriction enzymes. 5'-attggtaccttatggatccTTCAGGGCCAGAGACACCTG-3': DR-s1, 5'-aatagatctGACAAGGTGCCCCGAGTTGCGTGAG-3': DR-a1, 5'-aatcagatctACGC GAGACAGCAGGAGGGGGT-3': DR-a2.

DR-s1 is located from –1334 to –1314 upstream of the transcription start site for mouse *CYP1A1* gene (Fisher et al., 1990) and flanked with two restriction sites (*KpnI* and *BamHI*) at the 5' end. DR-a1 and DR-a2 are antisense primers and located from –989 to –962 and –844 to –822, respectively, and both them are flanked with a restriction site (*BglII*) at 5' end. The regulatory fragments amplified with DR-s1 and DR-a1, and with DR-s1 and DR-a2, were designated as DRE1 and DRE2, respectively. DRE1

has three Ah receptor recognition sequences. DRE2 has four Ah receptor recognition sequences and a GC box for signal enhancer (Fisher et al., 1990). DRE1 and DRE2 were digested with *KpnI* and *BglII*. DRE2 was cloned into *KpnI*–*BamHI* site of pUC-18 (TAKARA) to create pUC-18 DRE2. Digested DRE1 was ligated into *KpnI* and *BamHI* sites of pUC-18 DRE2 to create pUC-18 DRE1/2, and then digested with *BamHI* and *Sall* to obtain the fragment of about 950 bp containing DRE1 and DRE2.

We constructed a new reporter plasmid for the AhR-mediated transcriptional assay using pIND-MCS-LUC, based on the ecdysone-inducible mammalian expression vector pIND/Hygro harboring hygromycin resistant gene (Invitrogen), the construction method of pIND-MCS-LUC being previously described (Kojima et al., 2003). First, pIND-MCS-LUC was digested with *BanII* followed by blunting treatment with T4 DNA polymerase, and digested with *HindIII*. Next, the fragment containing luciferase gene (*luc*⁺) and SV40 late poly(A) signal was obtained from pGL3-Basic Luciferase Reporter Vector (Promega) by digesting with *Sall*, blunting with Klenow enzyme, and digesting with *HindIII*. Then, this fragment was ligated into the pIND-MCS-LUC already digested as mentioned above to create pIND-MCS-LUC2. Finally, the pUC-18 DRE1/2 fragment (about 950 bp) obtained as mentioned above was inserted into *BglII* and *XhoI* sites of the pIND-MCS-LUC2, and designated as pIND-GCDR7 (Fig. 1a), which contained seven DREs sequences and one GC box. After each recombinant DNA treatment, the arrangement of the insert was verified by DNA sequencing.

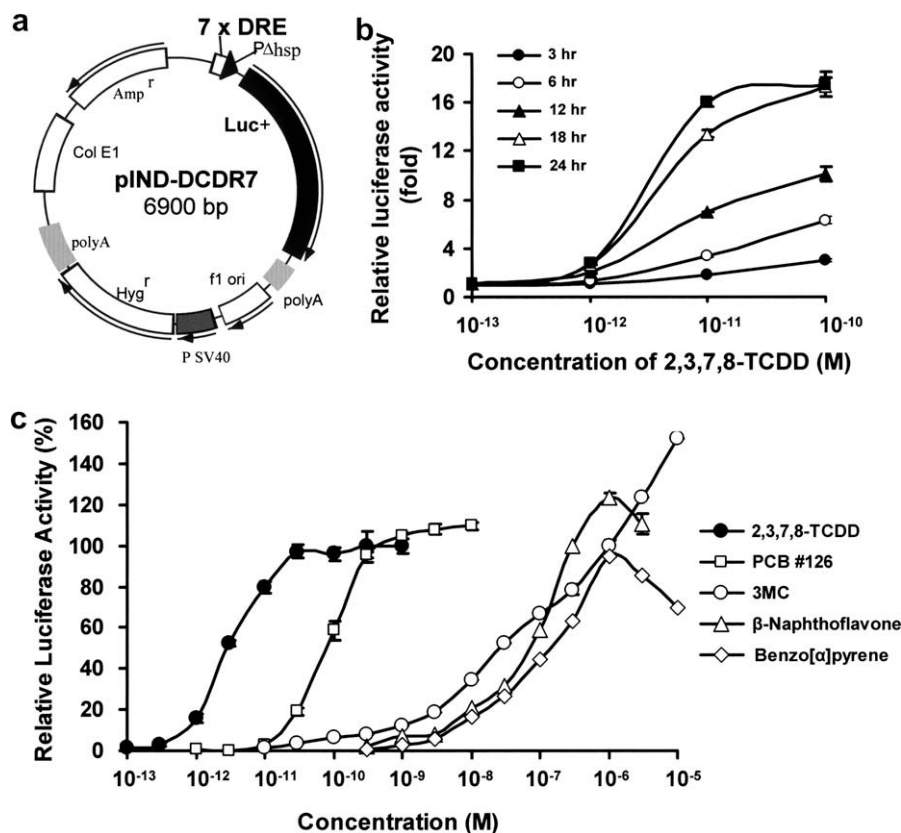


Fig. 1. Luciferase reporter gene assay using DR-EcoScreen cells stably transfected with pIND-DCDR7 plasmid. (a) Structure of the dioxin-responsive luciferase reporter plasmid, pIND-DCDR7. (b) Effects of incubation time on induction of luciferase reporter gene expression by TCDD in the DR-EcoScreen assay. DR-EcoScreen cells were treated with increasing concentrations of TCDD, and were assayed in different incubation time for 3–24 h. Values are fold-induction of the AhR agonistic activity over the vehicle control (means $\pm$ SD of three independent experiments). (c) Dose–response curves of AhR ligands in the DR-EcoScreen assay. DR-EcoScreen cells were treated with increasing concentrations of TCDD, PCB # 126, 3 MC, β -naphthoflavone, and benzo[α]pyrene to detect AhR agonistic activity. The luciferase activity was expressed as percentage induction, with 100% activity defined as the activity achieved with 10^{–10} M TCDD for AhR. Values represent the means $\pm$ SD of three independent experiments performed in duplicate.

2.4. Establishment of DR-EcoScreen cells and cell culture condition

About 16 h prior to the transfection, Hepa1c1c7 cells were seeded at a confluency of 50% in a 6-well plate in 2 mL of culture medium (α -MEM with 5%FBS) per well. Transfection was carried out with FuGENE[®]6 Transfection Reagent according to the instruction of the manufacturer. Briefly, 1 μ g of pIND-GCDR7 and 3 μ L of FuGENE[®]6 Transfection Reagent were added to each well. After 24 h incubation, the cells in each well were trypsinized, resuspended in the culture medium containing 150 μ g mL⁻¹ of hygromycin, and divided equally to plate in two 100 mm petri dishes. The culture medium was replaced every four d until colonies became available for isolation (for about two weeks).

After cells were exposed to 20 pM of TCDD for 24 h, luciferin (final concentration: 20 nM) was added into a 100 mm petri dish. After further incubation for 5 min, luminescence image was scanned through a photon detecting charge-coupled device (CCD) camera (Night OWL, Perkin–Elmer), which took 10 min per dish. Each luciferase positive clone was isolated using a cloning ring and individually cultured in a well of a 24-well plate. After proliferation, each clone was trypsinized, resuspended, seeded a 1/10 volume into a 100 mm petri dish, and further cultured. With repeated these steps, the clone having the highest luciferase activity and the greatest fold of induction ratio to background luciferase activity was selected, and named as DR-EcoScreen.

2.5. Luciferase reporter gene assay using DR-EcoScreen cells

The DR-EcoScreen cell line was maintained in α -MEM supplemented with 5% FBS, antibiotics and glutamine at 37 °C in an atmosphere of 5% CO₂/95% air under saturating humidity, and passaged twice every week by trypsinization with 0.25% trypsin/0.02% EDTA. For screening assay, cells were trypsinized and suspended at a density of 1.0×10^5 cells mL⁻¹ in α -MEM containing 5% of CD-FBS. Ninety μ L of the cell suspension was seeded in a well of a 96-well white bottom plate (#136102 Nunclon[™], Nalge Nunc, Denmark) (final density: 9000 cells well⁻¹). After 24 h cultivation at 37 °C, 10 μ L of various concentrations of chemicals that were dissolved in 1% DMSO was added to each well (final concentration of DMSO was 0.1%). Following further 24 h of cultivation, 100 μ L of Steady-Glo[™] reagent was added to each well. After the plate was shaken at room temperature for five min, the luminescence was measured with a microplate-luminometer (Wallac 1420 ARVO[™] SX, Perkin–Elmer).

2.6. Cytotoxicity test and evaluation of AhR agonistic activity

DR-EcoScreen cells were plated and cultured in the same way as the luciferase reporter gene assay, and added alamar blue dye instead of Steady-Glo[™] reagent. The cytotoxicity was measured in accordance with the manufacturer's protocol of alamarBlue[™] (AbD Serotec, Kidlington, UK) (O'Brien et al., 2000). This is a colorimetric assay to measure the metabolic activity of living cells based on the conversion of alamar blue dye (resorufin, blue) into resorufin (red). Absorbance was read at a test wavelength of 575 nm and a reference wavelength of 610 nm with a microplate-spectrophotometer, MPRA4i (TOSOH Co., Ltd., Osaka, Japan).

To estimate the potency of receptor–agonistic activity of the pesticide tested, its concentrations exhibiting the response equal to 50% and 20% of the maximal response of 10^{-10} M TCDD for AhR were evaluated from a dose–response curve of the luminescence intensity, and expressed as 50% and 20% relative effective concentration, REC₅₀ and REC₂₀, respectively. When the activity of the test compound was higher than REC₂₀ within the concentration tested ($\sim 10^{-7}$ – 10^{-5} M), we judged the pesticide to be positive for activity.

2.7. Animal experiments, isolation of RNA and reverse transcription

Six-week-old female C57BL/6 mice (18–20 g) were purchased from Japan SLC Inc. (Shizuoka, Japan) and maintained under controlled environmental conditions with a 12 h light/12 h dark cycle. Food and tap water were available *ad libitum*. Mice were intraperitoneally injected with 30, 100 or 300 mg kg⁻¹ body weight of the pesticide prepared in corn oil, or 100 mg kg⁻¹ body weight of 3 MC as a positive control, and sacrificed 14 h later. Control mice received corn oil alone. The isolation of RNA from the liver and reverse transcription of RNA was performed as previously described (Takeuchi et al., 2006). The mice were cared for in accordance with “The Regulations of Animal Experimentation of Hokkaido Institute of Public Health”, which are based on “The Guidelines of the International Committee on Laboratory Animals, 1974”.

2.8. Real-time quantitative PCR

We measured CYP1A1 and CYP1A2 mRNA levels in the liver of mice treated with pesticides or a vehicle by real-time quantitative PCR method previously reported (Takeuchi et al., 2006). The following primers and probes for detection were designed by Bioscience Technologies (Novato, CA, USA): for CYP1A1, 5'-AGATTTTGTTTTCTTTGGGAGGAA-3' and 5'-GGCTGGAAGTCCATACATGGA-3' as primers, and 5'-FAM d(AAGGGCATAGGCAGCCACCTAGATCATG) BHQ-1-3' as a probe; for CYP1A2, 5'-AGGAGCGCTGTATCTA CATAACCA-3' and 5'-GCGGAACACAAAGGGGTCT-3' as primers, and 5'-FAM d(CAGGTCAACCATGATGAGAAGCAGTGGA) BHQ-1-3' as a probe; for β -actin, 5'-AAAGGCTAATGGTGCCTACAGATT-3' and 5'-TGGAAACCATGGCTGTCCAT-3' as primers, and 5'-FAM d(AGCTCCTTGGATTGGGTATGGTTTGCTCTT) BHQ-1-3' as a probe. The amounts of CYP1A1 and CYP1A2 mRNAs were normalized to that of β -actin mRNA.

2.9. Data analysis

An analysis of variance (ANOVA) followed by Bonferroni correction was used to evaluate the differences in CYP1A1 and CYP1A2 mRNA levels determined by TaqMan[®] (Applied Biosystems) assay between the control group and each of the chemical groups. Statistical significance was set at $p < 0.05$. Data were presented as the means $\pm$ SD of five mice per group.

Table 2
Comparison of AhR agonistic activities of five AhR ligands and eleven pesticides

Compound	REC ₅₀ (M) ^a	REC ₂₀ (M) ^a	Relative potency ^b
TCDD	2.8×10^{-12}	1.6×10^{-12}	1
PCB # 126	7.6×10^{-11}	3.3×10^{-11}	0.048
3-MC	2.6×10^{-8}	2.0×10^{-9}	0.0008
β -Naphthoflavone	7.3×10^{-8}	9.4×10^{-9}	0.00017
Benzo[<i>a</i>]pyrene	1.3×10^{-7}	1.6×10^{-8}	0.0001
Prochloraz	N.D. ^c	7.1×10^{-7}	2.3×10^{-6}
Diuron	2.9×10^{-6}	8.7×10^{-7}	1.8×10^{-6}
Linuron	2.3×10^{-6}	9.3×10^{-7}	1.7×10^{-6}
Propanil	7.2×10^{-6}	1.1×10^{-6}	1.5×10^{-6}
Isoxathion	N.D.	2.0×10^{-6}	8.0×10^{-7}
Diethofencarb	N.D.	2.6×10^{-6}	6.1×10^{-7}
Bifenox	N.D.	3.6×10^{-6}	4.4×10^{-7}
Acifluorfen-methyl	N.D.	3.7×10^{-6}	4.3×10^{-7}
Quinalpho	N.D.	4.0×10^{-6}	4.0×10^{-7}
Chlorpropham	N.D.	5.0×10^{-6}	3.2×10^{-7}
Chlorpyrifos	N.D.	6.5×10^{-6}	2.5×10^{-7}

^a REC₅₀ and REC₂₀ are represented as the concentration of the test compound showing 50% and 20% of the agonistic activity of 10^{-10} M TCDD, respectively.

^b Relative potency is expressed as a hundredfold ratio of REC₂₀ of TCDD to that of a test compounds; i.e., the relative potency of TCDD is arbitrarily set at 1.

^c Not detectable.

Fig. 2. Agonistic effects of 200 pesticides in the DR-EcoScreen assay. DR-EcoScreen cells were treated with 200 pesticides to detect AhR agonistic activity. To avoid cytotoxic reaction due to pesticides, assays were performed for chlorothalonil and fluzainam at concentrations of $\leq 3 \times 10^{-6}$ M, and for other pesticides at concentrations of $\leq 10^{-5}$ M. The luciferase activity was expressed as percentage induction, with 100% activity defined as the AhR agonistic activity achieved by 10^{-10} M TCDD. Values represent the means $\pm$ SD of three independent experiments performed in duplicate.

As shown in Fig. 1b, the luciferase activity induced by TCDD increased in a dose-dependent manner at concentrations from 10^{-13} M to 10^{-10} M, and incubation time-dependently from 3 h to 24 h. Its activity was detectable even at a very low concentration of less than 1×10^{-12} M for incubation time of 12–24 h. The minimal detection limit (MDL; means of control solvent + $10 \times$ SD, $n = 8$) for TCDD was estimated to $0.003 \text{ pg well}^{-1}$ (1×10^{-13} M). The maximal intensity of the AhR activity induced by 1×10^{-10} M TCDD was achieved at 18–24 h incubation time, exhibiting approximately 17-fold that of control solvent (0.1% DMSO).

3.2. Responses of AhR ligands in the DR-EcoScreen assay

The DR-EcoScreen cells were treated with well-known AhR ligands such as TCDD, PCB # 126, 3-MC, β -naphthoflavone, and benzo[α]pyrene. Fig. 1c shows that these chemicals dose-dependently transactivated the receptor even at very low concentrations. From these dose-response curves, both REC_{50} and REC_{20} values of five AhR ligands for AhR were respectively described in Table 2. The relative potency calculated from the REC_{20} values indicates that the AhR agonistic activity of PCB # 126, 3 MC, β -naphthoflavone, and benzo[α]pyrene were about 21, 1250, 5900 and 10000-fold smaller than that of TCDD, respectively. Thus, a rapid and simple assay system using DR-EcoScreen cells is highly sensitive and reasonable to detect TCDD and related AhR ligands.

3.3. Transcriptional activity of 200 pesticides in DR-EcoScreen assay

We examined AhR agonistic activity of 200 pesticides at concentrations from 1×10^{-7} to 1×10^{-5} M by the *in vitro* reporter gene assay using DR-EcoScreen cells. Fig. 2 shows the screening result of AhR-mediated transcriptional activity induced by the 200 pesticides (1×10^{-5} M). We found that eleven pesticides, which included two diphenyl ether-type herbicides (acifluorfen-methyl and bifenox), three organophosphorus insecticides (chlorpyrifos, isoxathion, and quinalphos), two carbamate pesticides (chlorpropham and diethofencarb), an acid amide-type herbicide (propanil), two urea-type herbicides (diuron and linuron), and a imidazole fungicide (prochloraz) induced AhR-mediated transcriptional activity that is larger than 20% of the 10^{-5} M TCDD-induced maximal activity. On the other hand, organochlorine-, pyrethroid-, and triazine-type pesticides tested, showed little AhR agonistic activity, and several pesticides such as carbaryl, ethoxyquin and thiophanate-methyl, showed slightly weak agonistic activity. Most of the pesticides used were not shown to be cytotoxic to the DR-EcoScreen cells at the concentration of 1×10^{-5} M by the cytotoxicity test with the Alamar Blue kit except for chlorothalonil and fluazinam, which were cytotoxic above 3×10^{-6} M (data not shown).

Fig. 3 shows the dose-dependency of AhR transactivation induced by above-mentioned eleven pesticides. Among these pesticides, propanil, diuron, linuron and prochloraz, showed AhR agonistic activity in a lower concentration than other seven pesticides

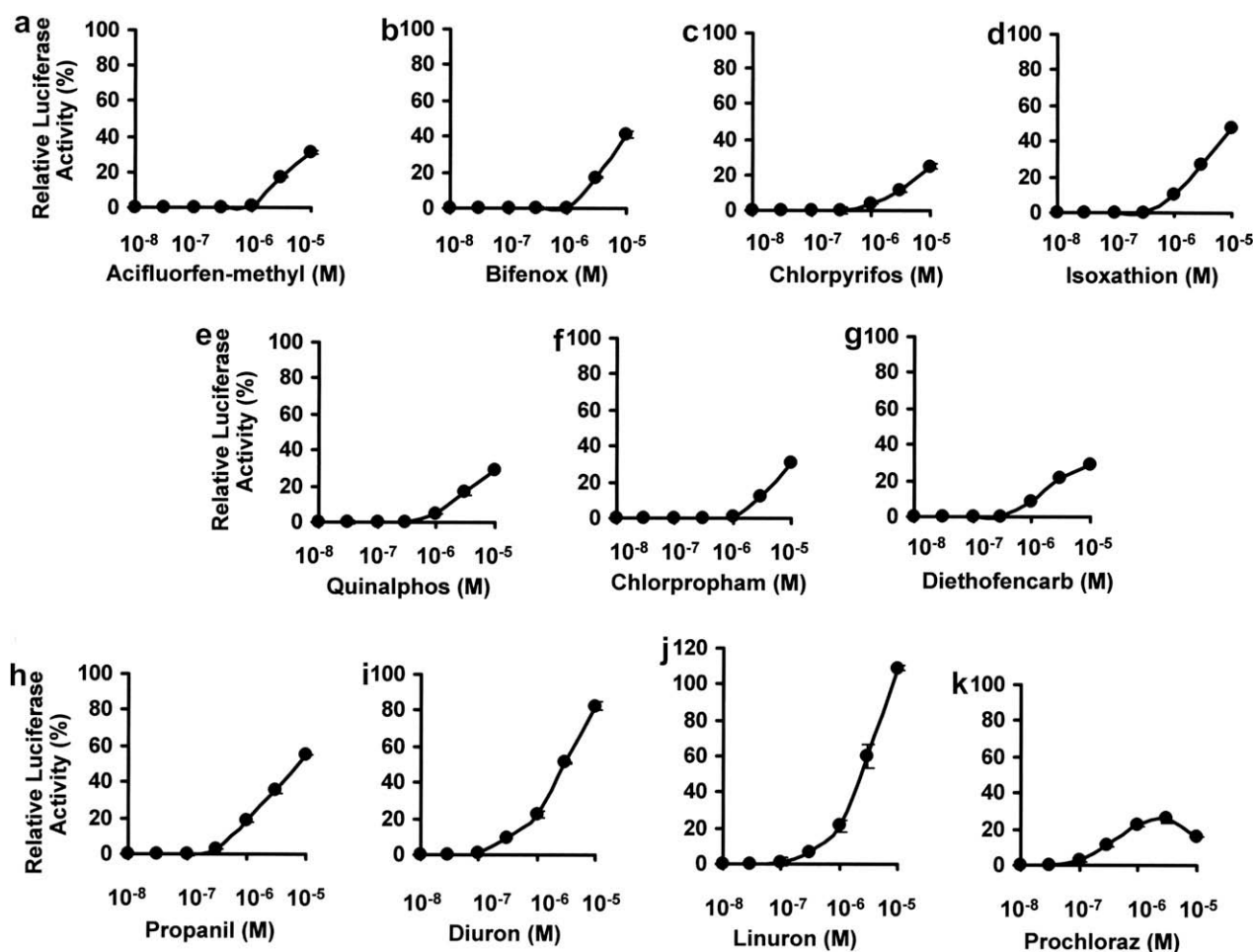


Fig. 3. Dose-response curves of 11 pesticides showing AhR agonistic activity. DR-EcoScreen cells were treated with increasing concentrations of acifluorfen-methyl (a), bifenox (b), chlorpyrifos (c), isoxathion (d), quinalphos (e), chlorpropham (f), diethofencarb (g), propanil (h), diuron (i), linuron (j), and prochloraz (k) to detect AhR agonistic activity. The luciferase activity was presented as percentage induction, with 100% activity defined as the activity achieved with 10^{-10} M TCDD for AhR. Values represent the means $\pm$ SD of three independent experiments performed in duplicate.

(Figs. 3h–k), although the AhR agonistic activity of prochloraz showed the slight repression at a concentration of 1×10^{-5} M. Table 2 shows the REC₅₀ and/or REC₂₀ values of the eleven AhR-agonistic pesticides; the potencies (REC₂₀ values) decreased in the order, prochloraz > diuron > linuron > propanil > isoxathion > diethofencarb > bifenoxy, acifluorfen-methyl > quinalphos > chlorpropham > chlorpyrifos. The AhR agonistic activities of these pesticides were 500000-fold weaker than that of TCDD. On the other hand, chlorpyrifos-oxon and isoxathion-oxon, which are the metabolites of chlorpyrifos and isoxathion, respectively, had no effect on luciferase induction in the DR-EcoScreen cells (data not shown).

3.4. Effect of propanil, diuron, and linuron on *in vivo* expression of CYP1A1 and CYP1A2

Since induction of CYP1A1 and CYP1A2 was speculated to be an early and perhaps essential step in the AhR signaling pathway (Dey et al., 1999; Kimura et al., 1986), we examined whether three herbicides (propanil, diuron, and linuron) showing high AhR agonistic activity *in vitro* (Fig. 3, Table 2) affect liver CYP1A levels *in vivo*. Total liver RNA was isolated from C57BL/6 mice intraperitoneally injected with propanil, diuron, or linuron (30, 100, and 300 mg kg⁻¹), and the expression levels of CYP1A1 and CYP1A2 mRNAs were assessed by a real-time quantitative RT-PCR system. As shown in Fig. 4a, only the treatment with the highest dose (300 mg kg⁻¹) of these herbicides induced significant amounts of CYP1A1 mRNAs compared with the control group, in spite of about half as much as the amount induced by a positive control, 3 MC (100 mg kg⁻¹). On the other hand, the expression levels of CYP1A2 mRNAs were significantly and dose-dependently increased by injection of these three herbicides over 100 mg kg⁻¹ (Fig. 4b), though they were also less than half the level induced by 100 mg kg⁻¹ of 3 MC.

4. Discussion

The AhR is a ligand-dependent transcription factor, which almost every vertebrate has, and mediates many of the biological and toxic actions induced by chemicals with structural diversity including environmental contaminants such as TCDD (Hankinson, 1995; Denison and Whitlock, 1995). Therefore, the development of *in vitro* screening assay of environmental chemicals having potentially harmful biological activities is extremely crucial to en-

sure intrinsic environmental safety. In this study, we have developed DR-EcoScreen, a cell line that stably expresses AhR responsive luciferase reporter gene construct. The reporter gene assay using DR-EcoScreen cells could detect the luciferase activity induced by low concentrations of well-known AhR agonists. In particular, TCDD was detected to induce AhR agonistic activity even at a very low concentration of 1×10^{-13} M (MDL), and DR-EcoScreen cell line was thought to be more sensitive to TCDD than other recombinant reporter gene cell bioassays described by Nagy et al. (2002). This advantage of DR-EcoScreen cells may be due to the difference that repeat number (seven copies) of DRE in this cell line is larger than those in other reporter cell lines. On the other hand, the potencies of AhR activity induced by other AhR agonists were 10–1000 fold smaller than that of TCDD, which corresponded to the results previously reported (Garrison et al., 1996). Thus, this rapid and sensitive DR-EcoScreen assay system has potential usefulness for screening a large number of environmental chemicals for AhR agonistic activity, and is compatible with high-throughput automation.

By DR-EcoScreen assay we characterized the AhR agonistic activity of 200 pesticides, all of which had been already screened for ER α , ER β , AR, PPAR α , and PPAR γ agonistic activities (Kojima et al., 2004; Takeuchi et al., 2006). These pesticides tested were selected according to the frequency of usage in Japan and other countries. In this study, we estimated the AhR agonistic activity using not only REC₅₀, which is more commonly used, but also REC₂₀ because of screening for weak AhR agonists such as nonclassical AhR ligands. The usage of REC₂₀ makes possible the comparison of potency between AhR agonistic pesticides and TCDD, and the judgment that pesticide is positive or negative as an AhR agonist. Here, we have found that eleven pesticides have AhR agonistic activity *in vitro* (Figs. 2 and 3). Of these eleven pesticides, chlorpyrifos, diuron and prochloraz have been already reported to be AhR agonists by several studies using various kinds of stable cell lines. Long et al. (2003) performed transactivation assay using the human hepatoma TV101L cell line and rat hepatoma H4IIE cell line to find that chlorpyrifos, prochloraz and iprodione may be an AhR agonist, and especially, prochloraz have AhR agonistic activity above the concentration of 1×10^{-6} M. In addition, they also reported that prochloraz may be both AhR agonist and antagonist in an assay using TV101L cells (Long et al., 2003). This suggests that in our assay the repressive effect by 1×10^{-5} M of prochloraz may

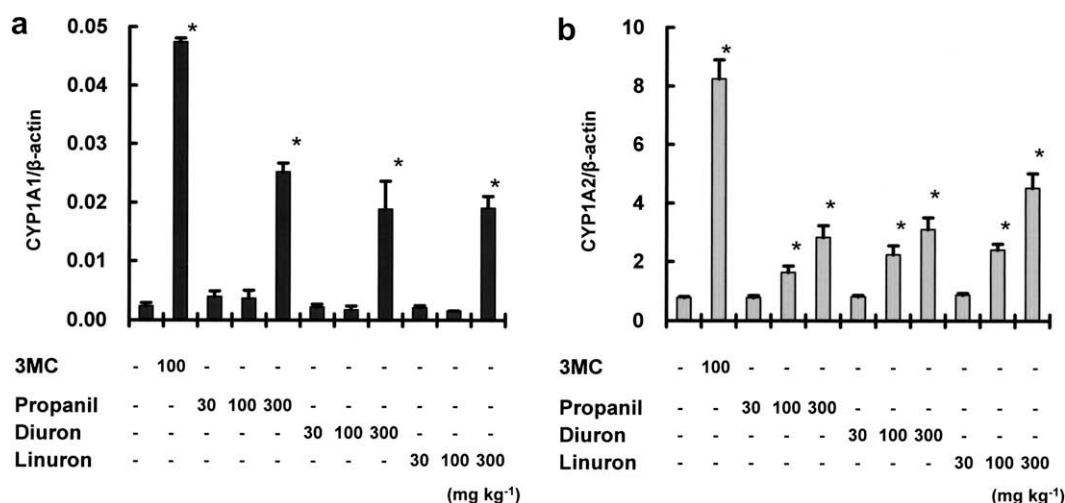


Fig. 4. Quantification of CYP1A1 and CYP1A2 mRNAs in the liver tissue of pesticide-treated mice. C57BL/6 mice were intraperitoneally injected with corn oil (vehicle control), 100 mg kg⁻¹ of 3 MC (positive control), 30, 100, or 300 mg kg⁻¹ of propanil, diuron and linuron, and sacrificed 14 h after the injection. CYP1A1 and CYP1A2 mRNA expressions of the individual mouse liver were amplified by quantitative RT-PCR and normalized to the expression of the housekeeping β-actin gene. Each determination was performed in triplicate for each cDNA. Values represent the means ± SD of five animals. *P < 0.05 vs. the vehicle control (leftmost bar).

be due to acting as an AhR antagonist. However, we found that a fungicide, iprodione, did not show any agonistic activity at concentrations from 1×10^{-7} M to 1×10^{-5} M (Fig. 2). These differences in response to chemicals between cell lines may be involved in differences in cell membrane transport, metabolic degradation processes, affinity for AhR, AhR/Arnt status, and other transcription co-factors of species and tissue used. Meanwhile, diuron was reported to have a possibility to interact with AhR of yeast, and AhR-dependent transcriptional activity of stable mammalian cell lines (Noguerol et al. 2006; Zhao et al. 2006). As a result, excluding these three pesticides (chlorpyrifos, diuron, and prochloraz), we first found other eight pesticides (acifluorfen-methyl, bifenox, isoxathion, quinalphos, chlorpropham, diethofencarb, propanil, and linuron) to possess AhR agonistic activity.

As shown in Fig. 5, the eleven pesticides were found to possess common chemical structures. Two diphenyl ether-type herbicides (acifluorfen-methyl and bifenox) have almost the same structure but a difference in residue; a trifluoromethyl residue in the former is replaced by a chlorine residue in the latter. Meanwhile, three organophosphorus insecticides, (chlorpyrifos, isoxathion, and quinalphos) have a common structure, a diethylthiophosphoryl residue $((\text{CH}_3\text{CH}_2\text{O})_2\text{-P(=S)-O-})$ that binds to an aromatic heterocycle ring at the carbon atom next to the nitrogen atom. However, chlorpyrifos-oxon and isoxathion-oxon, which belong to the same organophosphorus insecticide group but have a diethoxyphosphoryl residue $((\text{CH}_3\text{CH}_2\text{O})_2\text{-P(=O)-O-})$ instead of a

diethylthiophosphoryl residue differing in only one atom, showed little AhR-mediated effects (data not shown). This indicated that these two compounds might be metabolically oxidized in the environment or body so that their daughter compounds lost AhR agonistic activities. Two carbamate-type pesticides (chlorpropham and diethofencarb) have a common structure and differ in substituent bound to the benzene ring, a chlorine residue for chlorpropham and two ethoxy residues for diethofencarb. This suggests that the common structure, *N*-phenyl *O*-isopropyl carbamate, is attributed to their agonistic activities via AhR. An acid amide-type herbicide (propanil) and two urea-type herbicides (diuron and linuron) have a common chemical structure: a 3,4-dichlorophenyl residue binds to a nitrogen atom of the amido residue. Interestingly, some part of the structure of these three herbicides is similar to that of two above-mentioned carbamates-type insecticides, chlorpropham and diethofencarb. This partial common structure of these five pesticides may induce AhR agonistic activity. Finally, an imidazole fungicide prochloraz seems to be structurally similar to acifluorfen-methyl and bifenox.

Of 200 pesticides, three herbicides, propanil, diuron, and linuron, showed much more potent AhR agonistic activity in a dose-dependent manner than other pesticides. Zhao et al. (2006) demonstrated by a binding assay and reporter gene assays using stably transfected mouse, rat, guinea pig, and human cell lines that diuron and related substituted phenylureas interact with AhR. To examine whether propanil, diuron and linuron can induce AhR

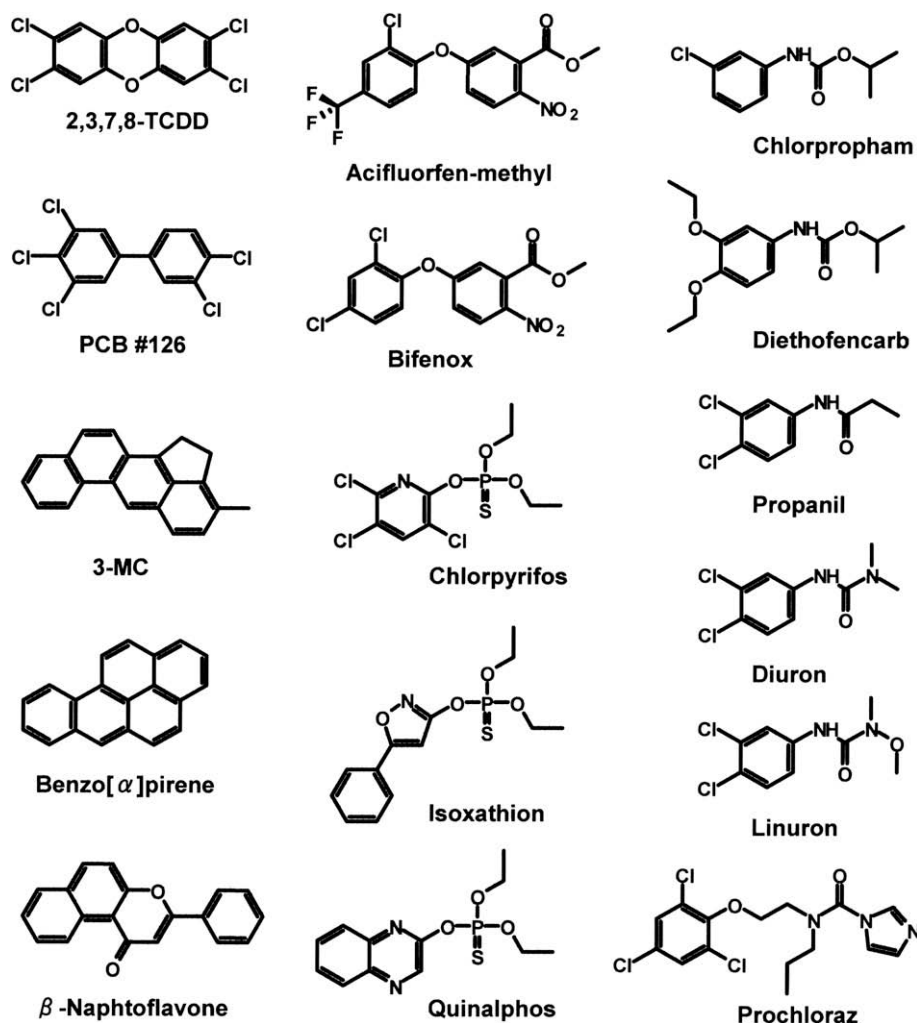


Fig. 5. Chemical structures of five AhR ligands and eleven pesticides having AhR agonistic activity.

agonistic activity *in vivo*, we measured transcription levels of the liver *CYP1A* subfamily genes, direct target genes of AhR (Honkakoski and Negishi, 2000), by quantitative RT-PCR analysis to reveal that they dose-dependently induced *CYP1A* mRNAs *in vivo* at the doses from 30 mg kg⁻¹ to 300 mg kg⁻¹ (Fig. 4). Although AhR binding assay was not performed in the present study, two herbicides (propanil and linuron), which we first screened *in vitro*, might also be AhR agonistic ligands *in vivo* as well as diuron. Schocket and Vincze (1990) demonstrated that diuron induced *CYP1A1*-dependent EROD activity in the rat, which also supports our results that these three herbicides can act as AhR agonists *in vivo*. Taken together, these results suggest that DR-EcoScreen assay system is useful in identifying AhR agonists among a large number of xenobiotic environmental chemicals.

Our finding suggests that many of pesticides (189 of 200) do not have AhR agonistic activity, and AhR agonistic pesticides may be small in number compared with pesticides possessing ER and/or AR activities. We previously demonstrated that 88 of 200 pesticides have estrogenic, anti-estrogenic and/or anti-androgenic activities through ER α , ER β and AR (Kojima et al., 2004): chlorpyrifos and isoxathion are estrogenic via ER α ; acifluorfen-methyl, bifenoxy, propanil, diuron, linuron, and prochloraz are antiandrogenic via AR; and it is interesting that quinalphos acts as both ER α / β agonist and AR antagonist. In addition, Bauer et al. (1998) presented evidence that propanil, diuron and linuron can bind to AR. Recently, it has been reported that AhR ligands modulate ER and AR signaling pathways both positively and negatively (Carlson and Perdew, 2002; Lin et al., 2002; Brunnberg et al., 2003; Matthews et al., 2005; Beischlag and Perdew, 2005). Notably, Ohtake et al. (2007) has demonstrated that liganded AhR acts as an E3 ubiquitin ligase, and degrades ER and AR not only *in vitro* but also *in vivo*. Thus, these results and our present study taken together suggest that AhR agonistic pesticides may act as not only AhR agonists but also chemicals, which affect on functions of ERs and AR through direct receptor binding or multiple indirect receptor pathways via AhR.

To date, a number of studies have demonstrated that cell-based bioassay systems such as CALUX assay can be possible alternative screening assays of dioxins (polychlorinated dibenzo-*p*-dioxins (PCDDs), dibenzofurans (PCDFs), and dioxin-like PCBs) to high-resolution gas chromatography/high-resolution mass spectrometry (HRGC/HRMS), which requires expensive equipment and highly trained analysts (Behnisch et al., 2002; Murk et al., 1996; Tsutsumi et al., 2003). Therefore, this sensitive, rapid and simple DR-EcoScreen assay is thought to be applicable to a simple monitoring system to determine dioxins and dioxin-like compounds in various field samples from air and water as well as foods. On the other hand, the physiological role of the AhR remains to be elucidated. Hahn (2002) reported that AhR is conserved throughout a number of animal species from invertebrates to vertebrates, which suggests that AhR plays fundamental and essential roles in some physiological processes. DR-EcoScreen cells may be also useful in clarifying physiological function of endogenous AhR ligands.

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